DATA606-Final Project proposal

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2026-11-08

```
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(lubridate)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(knitr)
```

Data Preparation

This dataset relevant to Worldwide natural disaster occurences documented between 2023 and 2025. Recent worldwide natural disaster events like earthquake, hurricane, wildfire etc. really concern me and I am excited to work on this dataset. This dataset contains important characteristics such the type of disaster, location, degree of severity, number of victims, financial losses, and emergency response timings are all captured here and also gives an opportunity to working with enormous numerical data.

I will identify important patterns, trends, and connections across many disaster categories using structured exploratory data analysis (EDA). Also get insights information through data driven approach and develop more visualize data to improve emergency response plans, risk mitigation techniques and disaster preparedness. So data driven insights are essential for identifying susceptible areas and allocating resources as emergency needed.

Research question

- Which disaster types occur most frequently?
- How does the distribution of economic loss differ across disaster types?
- Is higher severity associated with greater estimated economic loss?
- How are response times related to the affected population?
- Where are the synthetic events geographically concentrated?
- How do major-disaster rates and infrastructure damage differ by disaster type?

Cases

Here, 20000 natural events happened for 7 types of disasters across different country in world wide.

Data Collection

Data download from Kaggle and then uploaded to Github. I am reading the .csv from github raw form.

Type of study

This is an experimental study for worldwide natural disaster occurrences documented between 2023 and 2025.

Data Source

I obtained the data from - `https://github.com/CandySoftInc/DATA606-Final-Project/blob/main/synthetic_disaster_`

Dependent Variable

Here "*severity_level*", "*affected_population*", "*estimated_economic_loss_usd*", "*infrastructure_damage_index*", "*response_time_hours*" & "*aid_provided*" data depends based on "*disaster_type*" & "*is_major_disaster*" data.

Here "*disaster_type*" & "*location*" value is qualitative and all other values are quantitative.

Independent Variable

Here two independent variables are - "*severity_level*" (quantitative) and "*disaster_type*" (qualitative)

Conclusion

The scope of this project and the datasets used allow for a fairly in depth analysis of the World wide natural disaster which documented between 2023 and 2025. This project will help me to learn, grow and develop my knowledge with subject "DATA606-Statistics and Probability" and on forward, give me a hand-zone opportunity to develop my skill sets with R programming.